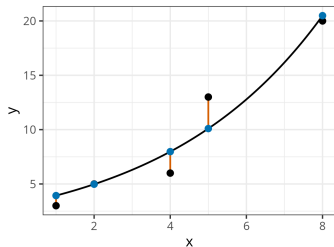
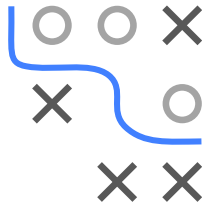


Optimization in Machine Learning

Second order methods

Gauss-Newton



Learning goals

- Least squares
- Gauss-Newton
- Levenberg-Marquardt

LEAST SQUARES PROBLEM

- Consider the problem of minimizing a sum of squares

$$\min_{\theta} g(\theta),$$

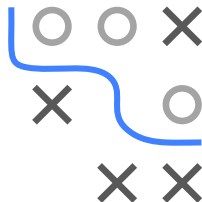
where

$$g(\theta) = r(\theta)^T r(\theta) = \sum_{i=1}^n r_i(\theta)^2$$

and

$$\begin{aligned} r : \mathbb{R}^d &\rightarrow \mathbb{R}^n \\ \theta &\mapsto (r_1(\theta), \dots, r_n(\theta))^T \end{aligned}$$

maps parameters θ to residuals $r(\theta)$



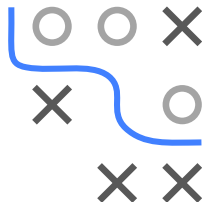
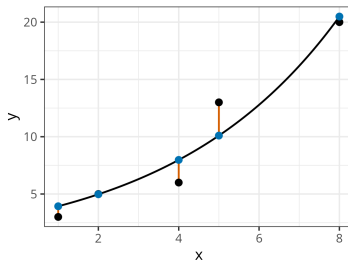
LEAST SQUARES PROBLEM

- Risk minimization with squared loss $L(y, f(\mathbf{x})) = (y - f(\mathbf{x}))^2$
- Least squares regression:

$$\mathcal{R}_{\text{emp}}(\theta) = \sum_{i=1}^n L(y^{(i)}, f(\mathbf{x}^{(i)} | \theta)) = \sum_{i=1}^n \underbrace{\left(y^{(i)} - f(\mathbf{x}^{(i)} | \theta)\right)^2}_{r_i(\theta)^2}$$

- $f(\mathbf{x}^{(i)} | \theta)$ might be a function that is **nonlinear** in θ
- Residuals: $r_i = y^{(i)} - f(\mathbf{x}^{(i)} | \theta)$
- **Example:**

$$\begin{aligned}\mathcal{D} &= \left((\mathbf{x}^{(i)}, y^{(i)}) \right)_{i=1, \dots, 5} \\ &= ((1, 3), (2, 5), (4, 6), (5, 13), (8, 20))\end{aligned}$$



LEAST SQUARES PROBLEM

- Suppose, we suspect an *exponential* relationship between $x \in \mathbb{R}$ and y

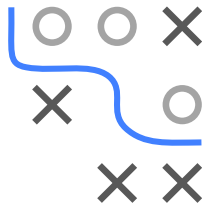
$$f(x | \theta) = \theta_1 \cdot \exp(\theta_2 \cdot x), \quad \theta_1, \theta_2 \in \mathbb{R}$$

- Residuals:**

$$r(\theta) = \begin{pmatrix} \theta_1 \exp(\theta_2 x^{(1)}) - y^{(1)} \\ \theta_1 \exp(\theta_2 x^{(2)}) - y^{(2)} \\ \theta_1 \exp(\theta_2 x^{(3)}) - y^{(3)} \\ \theta_1 \exp(\theta_2 x^{(4)}) - y^{(4)} \\ \theta_1 \exp(\theta_2 x^{(5)}) - y^{(5)} \end{pmatrix} = \begin{pmatrix} \theta_1 \exp(1\theta_2) - 3 \\ \theta_1 \exp(2\theta_2) - 5 \\ \theta_1 \exp(4\theta_2) - 6 \\ \theta_1 \exp(5\theta_2) - 13 \\ \theta_1 \exp(8\theta_2) - 20 \end{pmatrix}$$

- Least squares problem:**

$$\min_{\theta} g(\theta) = \min_{\theta} \sum_{i=1}^5 \left(y^{(i)} - \theta_1 \exp(\theta_2 x^{(i)}) \right)^2$$



NEWTON-RAPHSON IDEA

- **Approach:** Calculate Newton-Raphson update direction by solving:

$$\nabla^2 g(\theta^{[t]}) \mathbf{d}^{[t]} = -\nabla g(\theta^{[t]}).$$

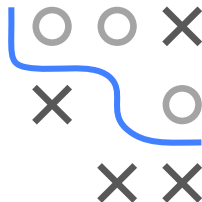
- Gradient is calculated via chain rule

$$\nabla g(\theta) = \nabla(r(\theta)^T r(\theta)) = 2 \cdot J_r(\theta)^T r(\theta),$$

where $J_r(\theta)$ is Jacobian of $r(\theta)$.

- In our example:

$$J_r(\theta) = \begin{pmatrix} \frac{\partial r_1(\theta)}{\partial \theta_1} & \frac{\partial r_1(\theta)}{\partial \theta_2} \\ \frac{\partial r_2(\theta)}{\partial \theta_1} & \frac{\partial r_2(\theta)}{\partial \theta_2} \\ \vdots & \vdots \\ \frac{\partial r_5(\theta)}{\partial \theta_1} & \frac{\partial r_5(\theta)}{\partial \theta_2} \end{pmatrix} = \begin{pmatrix} \exp(\theta_2 x^{(1)}) & x^{(1)} \theta_1 \exp(\theta_2 x^{(1)}) \\ \exp(\theta_2 x^{(2)}) & x^{(2)} \theta_1 \exp(\theta_2 x^{(2)}) \\ \exp(\theta_2 x^{(3)}) & x^{(3)} \theta_1 \exp(\theta_2 x^{(3)}) \\ \exp(\theta_2 x^{(4)}) & x^{(4)} \theta_1 \exp(\theta_2 x^{(4)}) \\ \exp(\theta_2 x^{(5)}) & x^{(5)} \theta_1 \exp(\theta_2 x^{(5)}) \end{pmatrix}$$



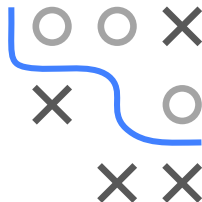
NEWTON-RAPHSON IDEA

- Hessian of g , $\mathbf{H}_g = (H_{jk})_{jk}$, is obtained via product rule:

$$H_{jk} = 2 \sum_{i=1}^n \left(\frac{\partial r_i}{\partial \theta_j} \frac{\partial r_i}{\partial \theta_k} + r_i \frac{\partial^2 r_i}{\partial \theta_j \partial \theta_k} \right)$$

- **But:**

Main problem with Newton-Raphson: Second derivatives can be computationally expensive.



GAUSS-NEWTON FOR LEAST SQUARES

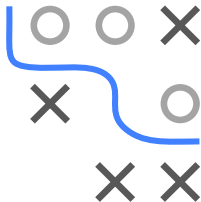
- Gauss-Newton approximates $\mathbf{H}_g$ by dropping its second order part:

$$\begin{aligned} H_{jk} &= 2 \sum_{i=1}^n \left(\frac{\partial r_i}{\partial \theta_j} \frac{\partial r_i}{\partial \theta_k} + r_i \frac{\partial^2 r_i}{\partial \theta_j \partial \theta_k} \right) \\ &\approx 2 \sum_{i=1}^n \frac{\partial r_i}{\partial \theta_j} \frac{\partial r_i}{\partial \theta_k} \\ &= 2 J_r(\theta)^T J_r(\theta) \end{aligned}$$

- **Note:** We assume that

$$\left| \frac{\partial r_i}{\partial \theta_j} \frac{\partial r_i}{\partial \theta_k} \right| \gg \left| r_i \frac{\partial^2 r_i}{\partial \theta_j \partial \theta_k} \right|.$$

- This assumption may be valid if: Residuals r_i are small in magnitude or functions are only “mildly” nonlinear s.t. $\frac{\partial^2 r_i}{\partial \theta_j \partial \theta_k}$ is small



GAUSS-NEWTON FOR LEAST SQUARES

- If $J_r(\theta)^T J_r(\theta)$ is invertible, Gauss-Newton update direction is

$$\begin{aligned}\mathbf{d}^{[t]} &= - \left[\nabla^2 g(\theta^{[t]}) \right]^{-1} \nabla g(\theta^{[t]}) \\ &\approx - \left[J_r(\theta^{[t]})^T J_r(\theta^{[t]}) \right]^{-1} J_r(\theta^{[t]})^T r(\theta) \\ &= -(J_r^T J_r)^{-1} J_r^T r(\theta)\end{aligned}$$

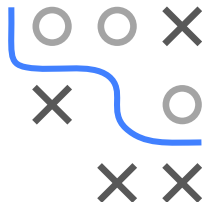
- **Advantage:**

Reduced computational complexity since no Hessian necessary.

- **Note:** Gauss-Newton can also be derived by starting with

$$r(\theta) \approx r(\theta^{[t]}) + J_r(\theta^{[t]})^T (\theta - \theta^{[t]}) = \tilde{r}(\theta)$$

and $\tilde{g}(\theta) = \tilde{r}(\theta)^T \tilde{r}(\theta)$. Then, set $\nabla \tilde{g}(\theta)$ to zero



LEVENBERG-MARQUARDT ALGORITHM

- **Problem:** Gauss-Newton may not decrease g in every iteration but may diverge, especially if starting point is far from minimum
- **Solution:** Choose step size $\alpha > 0$ s.t.

$$\mathbf{x}^{[t+1]} = \mathbf{x}^{[t]} + \alpha \mathbf{d}^{[t]}$$

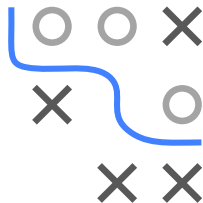
decreases g (e.g., by satisfying Wolfe conditions)

- However, if α gets too small, an **alternative** method is the

Levenberg-Marquardt algorithm

$$(J_r^T J_r + \lambda D) \mathbf{d}^{[t]} = -J_r^T r(\theta)$$

- D is a positive diagonal matrix
- $\lambda = \lambda^{[t]} > 0$ is the *Marquardt parameter* and chosen at each step



LEVENBERG-MARQUARDT ALGORITHM

- **Interpretation:** Levenberg-Marquardt *rotates* Gauss-Newton update directions towards direction of *steepest descent*
- Let $D = I$ for simplicity. Then:

$$\begin{aligned}\lambda \mathbf{d}^{[t]} &= \lambda (J_r^T J_r + \lambda I)^{-1} (-J_r^T r(\theta)) \\ &= (I - J_r^T J_r / \lambda + (J_r^T J_r)^2 / \lambda^2 \mp \dots) (-J_r^T r(\theta)) \\ &\rightarrow -J_r^T r(\theta) = -\nabla g(\theta) / 2\end{aligned}$$

for $\lambda \rightarrow \infty$

- **Note:** $(\mathbf{A} + \mathbf{B})^{-1} = \sum_{k=0}^{\infty} (-\mathbf{A}^{-1} \mathbf{B})^k \mathbf{A}^{-1}$ if $\|\mathbf{A}^{-1} \mathbf{B}\| < 1$
- Therefore: $\mathbf{d}^{[t]}$ approaches direction of negative gradient of g
- Often: $D = \text{diag}(J_r^T J_r)$ to get scale invariance (**Recall:** $J_r^T J_r$ is positive semi-definite $\Rightarrow$ non-negative diagonal)

